Classical Field Theory — Exercise Sheet 4

Formula, stability and weak derivatives

Solutions

Use the unitary Fourier transform $\widehat{f}(k) = (2\pi)^{-1/2}\int_{\mathbb{R}} e^{-ikx}f(x)\mathrm{d}x$ and $\|f\|_{H^1}^2 = \int_{\mathbb{R}}(1+k^2)|\widehat{f}(k)|^2\mathrm{d}k$. All wave equations on this sheet have one spatial dimension and zero mass. The solution class is that of Theorem 4.3.

Exercise 4.1 (Two initial data and two travelling packets). Let $f, g \in C_c^\infty(\mathbb{R};\mathbb{R})$ be initial displacement and velocity.

(a) Derive d'Alembert's formula from Lecture 4's Fourier multipliers. Use an oriented integral when $t < 0$. Verify the equation and both data, and justify identifying the formula with the energy-class solution.

(b) Let F, G be smooth real functions supported in $[-1/4, 1/4]$ and strictly positive on its interior. Take

$$f(x) = F(x+3) + 2G(x-2), \qquad g(x) = -F'(x+3) + 2G'(x-2).$$

Find the solution. For $t \geq 0$, find the open time intervals on which an observer at $x = 0$ sees each contribution. Where and when do the packet centres meet? Find the corresponding visibility interval at that position.

Solution. Lecture 4 constructs the solution by $\widehat{v} = \cos(t|k|)\widehat{f} + S_t(|k|)\widehat{g}$. Since $\cos(t|k|) = \cos(tk)$, Fourier translation gives the displacement term $[f(x+t) + f(x-t)]/2$. The integral representation of the sine multiplier gives its velocity contribution as

$$\frac{1}{2}\int_0^t [g(x+s) + g(x-s)]\mathrm{d}s = \frac{1}{2}\int_{x-t}^{x+t} g(y)\mathrm{d}y.$$

The identity holds also for negative t with oriented integrals. Fourier inversion and the time integral can be interchanged here because the initial Fourier transforms are Schwartz functions.

Write the last contribution as w. The fundamental theorem of calculus gives

$$w_t = \frac{1}{2}[g(x+t)+g(x-t)], \qquad w_x = \frac{1}{2}[g(x+t)-g(x-t)], \qquad w_{tt} = w_{xx} = \frac{1}{2}[g'(x+t)-g'(x-t)].$$

The translated displacement term also solves $v_{tt} = v_{xx}$. At zero it equals f, while its time derivative vanishes; $w(0) = 0$ and $w_t(0) = g$. On each compact time interval all these functions have a common compact spatial support and are smooth. Dominated convergence therefore gives $v \in C(H^1) \cap C^1(L^2)$. The classical equation implies the weak equation by integration against compact tests, so Lecture 4's uniqueness applies.

For the specified data, the travelling field

$$v(t,x) = F(x+3-t) + 2G(x-2+t)$$

has exactly the required two initial values, and hence is that solution. At $x = 0$ its arguments lie in $(-1/4, 1/4)$ precisely for $11/4 < t < 13/4$ and $7/4 < t < 9/4$, respectively. Strict positivity gives nonzero contributions throughout those intervals, and smooth support gives

zero at their endpoints. The centres follow $x = t - 3$ and $x = 2 - t$. They meet at $t = 5/2$, $x = -1/2$. At this position the two arguments are $5/2 - t$ and $t - 5/2$, so both visibility intervals are $(9/4, 11/4)$. The linear sum remains valid while the packets overlap.

Exercise 4.2 (Hadamard instability for an elliptic Cauchy problem). For each positive integer n, seek a smooth function $u_n(x, y)$, periodic in $x \in \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, satisfying

$$\partial_y^2 u_n + \partial_x^2 u_n = 0, \qquad u_n(x, 0) = 0, \qquad \partial_y u_n(x, 0) = e^{-\sqrt{n}} \cos(nx).$$

Find such a solution by separation of variables. Prove that the initial velocity tends to zero in every fixed $C^r(\mathbb{T})$ seminorm, while $\|u_n(\cdot, y)\|_{L^2(\mathbb{T})} \to \infty$ for every fixed $y > 0$. What continuity claim does this disprove? Why does it not contradict Lecture 4?

Solution. For a mode $u_n = a_n(y) \cos(nx)$, the equation is $a_n'' - n^2 a_n = 0$. Its specified data give $a_n(y) = e^{-\sqrt{n}} \sinh(ny)/n$. Each derivative of order $j \le r$ of the initial velocity has supremum at most $n^j e^{-\sqrt{n}} \le n^r e^{-\sqrt{n}}$. For fixed r, its logarithm $r \log n - \sqrt{n}$ tends to $-\infty$. Thus all the fixed smooth seminorms tend to zero.

Since $\int_0^{2\pi} \cos^2(nx) \mathrm{d}x = \pi$ for $n \ge 1$,

$$\|u_n(\cdot, y)\|_2 = \frac{\sqrt{\pi}}{n} e^{-\sqrt{n}} \sinh(ny).$$

For fixed $y > 0$ and sufficiently large n, $\sinh(ny) \ge e^{ny}/4$. The logarithm of the resulting lower bound is $ny - \sqrt{n} - \log n + \log(\sqrt{\pi}/4)$, which tends to $+\infty$. Consequently a Cauchy solution assignment agreeing with these modes cannot be continuous at zero from the smooth-data topology to L^2 on any positive slice. This is instability even among data possessing explicit smooth solutions; no assertion of existence for all smooth data is needed. The equation here is elliptic: it has $+\partial_x^2$, whereas Lecture 4's wave equation is $\partial_t^2 - \partial_x^2 = 0$. Reversing wave time would not change that sign.

Exercise 4.3 (A corner and its weak derivatives ⋆). Let $f(x) = e^{-|x|}$. Compute its unitary Fourier transform and its H^1 norm. Identify its first weak derivative by testing against every $\zeta \in C_c^\infty(\mathbb{R})$, and calculate its second distributional derivative. Decide whether f belongs to the space

$$H^2(\mathbb{R}) = \left\{ h \in L^2(\mathbb{R}) : \int_{\mathbb{R}} (1 + k^2)^2 |\widehat{h}(k)|^2 \mathrm{d}k < \infty \right\}.$$

Distinguish the value assigned to a weak derivative at the single point zero from the existence of a classical derivative there.

Solution. The two half-line integrals give

$$\widehat{f}(k) = \frac{1}{\sqrt{2\pi}} \left(\frac{1}{1 - ik} + \frac{1}{1 + ik} \right) = \frac{\sqrt{2/\pi}}{1 + k^2}.$$

Hence the Fourier definition and $\int_{\mathbb{R}} (1 + k^2)^{-1} \mathrm{d}k = \pi$ give $\|f\|_{H^1}^2 = 2$. Direct integration also gives $\|f\|_2^2 = 1$.

For a compact smooth test ζ, integration by parts on both half-lines gives

$$\int_{\mathbb{R}} f\zeta' = \zeta(0) - \int_{-\infty}^{0} e^x \zeta \mathrm{d}x - \zeta(0) + \int_0^{\infty} e^{-x} \zeta \mathrm{d}x = -\int_{\mathbb{R}} h\zeta,$$

where $h = e^x$ for $x < 0$ and $h = -e^{-x}$ for $x > 0$. Thus h is the weak derivative. It belongs to L^2 and has norm squared 1, agreeing with the Fourier norm identity. Its value at zero may be chosen arbitrarily without changing this L^2 element.

Differentiating h distributionally retains its jump. More explicitly, the same half-line calculation gives

$$\langle h', \zeta \rangle = -\int_{\mathbb{R}} h\zeta' = \int_{\mathbb{R}} f\zeta - 2\zeta(0).$$

Therefore $f'' = f - 2\delta_0$. The Fourier H^2 integrand is the nonzero constant $2/\pi$, so its integral diverges. Finally the classical one-sided derivatives of f at zero are 1 and -1; changing a weak derivative's value at one point cannot make these agree.

Exercise 4.4 (The sharp massless sine bound). Choose a real even $\rho \in C_c^\infty((-1,1))$ with $\|\rho\|_2 = 1$. For $0 < \varepsilon \le 1$, let g_ε satisfy $\widehat{g}_\varepsilon(k) = \varepsilon^{-1/2}\rho(k/\varepsilon)$, and let v_ε have initial data $(0, g_\varepsilon)$.

(a) Prove that g_ε is real and normalized in L^2, and that for every $T > 0$,

$$\sup_{|t|\le T} \|v_\varepsilon(t) - tg_\varepsilon\|_2 \le \frac{T^3\varepsilon^2}{6}.$$

(b) For fixed real t, determine the operator norm on real $L^2(\mathbb{R})$ of the Fourier multiplier $S_t(|k|)$, where $S_t(0) = t$. Explain why conservation of massless energy cannot give a bound for $\|v(t)\|_2$ uniform in both time and all initial data with energy $1/2$.

Solution. The transform is real and even, so it satisfies $\widehat{g}_\varepsilon(-k) = \overline{\widehat{g}_\varepsilon(k)}$. Fourier inversion preserves reality. Substitution $k = \varepsilon q$ and Plancherel give

$$\|g_\varepsilon\|_2^2 = \int_{\mathbb{R}} \varepsilon^{-1}|\rho(k/\varepsilon)|^2 \mathrm{d}k = 1.$$

The inverse transform is Schwartz, so all these data are permitted by Lecture 4. Integrating $|1 - \cos s| \le s^2/2$ gives $|\sin z - z| \le |z|^3/6$ for real z. Thus

$$|S_t(|k|) - t| \le \frac{|t|^3k^2}{6}, \qquad \|(S_t(|k|) - t)\widehat{g}_\varepsilon\|_2 \le \frac{|t|^3\varepsilon^2}{6}\|\widehat{g}_\varepsilon\|_2.$$

This proves the compact-time estimate, including $k = 0$ by continuity. In particular the reverse triangle inequality gives $\big|\|v_\varepsilon(t)\|_2 - |t|\big| \le |t|^3\varepsilon^2/6$.

Lecture 4's bound $|S_t| \le |t|$ supplies an operator norm at most $|t|$. The normalized real test family just constructed gives the reverse bound by taking $\varepsilon \to 0$. Hence the norm is $|t|$ for every real t. The initial energy is $\|g_\varepsilon\|_2^2/2 = 1/2$. Given a proposed common displacement bound M, choose a time with $|t| > M$ and then choose ε small enough that the displacement norm exceeds M. This disproves a bound uniform over this energy level and all time. It does not assert unbounded growth for each fixed datum; the finite-time estimate of Lecture 4 remains valid.